

Dimitri Chrysafis

GPU, Graphics & Machine Learning Systems

dimitri.chrysafis@gmail.com | 650-305-9037 | github.com/DimitriChrysafis | dimitrichrysafis.github.io

PROFILE

Computer science student with a **4.0 GPA** building GPU-accelerated simulation, graphics, geometry, and ML systems—from mathematical models and low-level kernels to polished interactive tools. Open-source contributor and competitive endurance athlete.

EDUCATION

University of Wisconsin–Madison — B.S. Computer Science, **GPA: 4.0**

Expected May 2027

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, Swift, SQL, WGSL/GLSL

GPU & Systems: WebGPU, Metal, CUDA/MPS, shader programming, Git

ML & Geometry: PyTorch, CatBoost, NumPy, scikit-learn, Open3D, LoRA/QLoRA, quantization, adapter merging

OPEN-SOURCE EXPERIENCE

SimpleX Chat — Privacy-first messaging platform

Open-Source Contributor

- Improved multi-device synchronization in the client messaging pipeline, tracing message and state flow across linked devices and resolving reconciliation edge cases.
- Updated connection-lifecycle handling to keep device state coherent and reduce duplicate-delivery scenarios through disconnects, reconnects, and concurrent sync.

SELECTED PROJECTS

Interactive 3D Fluid Simulator

Live Demo | GitHub | WebGPU, WGSL, JavaScript

- Designed a browser-native MLS-MPM physics engine and GPU pipeline that simulates **400,000 particles** in real time using parallel particle-to-grid, grid-update, and grid-to-particle compute passes.
- Eliminated CPU-side particle copies with shared storage buffers and direct rendering; implemented 128-thread workgroups, atomic fixed-point accumulation, dynamic piston boundaries, turbulence pulses, and adaptive render density.

Volumetric Sphere Packing for Arbitrary 3D Meshes

GitHub | Technical Write-up | PyTorch, Open3D

- Built a geometry pipeline that packs non-overlapping tangent spheres inside arbitrary OBJ meshes using Möller–Trumbore parity tests, nearest-surface queries, global grid seeding, and stochastic tangent growth.
- Vectorized containment, clearance, and overlap checks as batched PyTorch operations on CUDA/MPS; combined binary-search radius fitting, mesh decimation, surface sampling, and live Open3D visualization.

Tennis Match Prediction System

CatBoost, Python, scikit-learn

- Trained a gradient-boosted classifier on **92,000 ATP matches (1982–2024)**, engineering player and surface Elo, head-to-head history, recent form, and schedule-derived fatigue features.
- Reached **83.7% prediction accuracy**, outperforming baseline models by **12%**, through feature iteration and ensemble-based modeling of nonlinear player, surface, and recency effects.

Real-Time Fractal Visualization Engine

Demos & Write-up | WebGL, GLSL, Swift/Metal

- Implemented interactive Mandelbrot, Julia, Newton, and Kleinian renderers with complex recurrences and Möbius transformations evaluated per pixel in GPU fragment shaders.
- Added smooth pan/zoom, animated parameter interpolation, supersampling, and iteration-based coloring; also built a native Julia-set renderer in Swift/Metal for Apple GPUs.

Fourier Drawing Machine

GitHub | Demo | Python, NumPy, JavaScript

- Built a DFT pipeline that converts sampled or image-derived paths into complex coefficients, reconstructs them with rotating epicycles, and exports eased frame-by-frame animations; later moved computation to a web worker for an interactive browser version.

Interests: Competitive marathon running and triathlon; calisthenics and squash.